

Azure Honeypot & Threat Detection

SOC Lab Project Documentation

Project Summary

A Windows Virtual Machine was deliberately deployed on Azure with RDP port 3389 exposed to the internet as a honeypot. Microsoft Sentinel was configured to ingest Windows Security Event logs, detect successful RDP sign-ins, and hunt real-world brute-force attack traffic — all within hours of exposure.

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Date	May 20, 2026
Platform	Microsoft Azure
Tools Used	Microsoft Sentinel · Microsoft Defender for Cloud · KQL
VM	labtestvm · Windows · Standard D2s v3 · South Africa North

1. Project Overview

This project demonstrates a realistic Security Operations Center (SOC) workflow using cloud-native Azure tools. By intentionally exposing an RDP endpoint to the public internet, the lab simulates the threat surface of a misconfigured or poorly secured cloud workload. The goal was to observe real attack behavior, validate detection logic in Microsoft Sentinel, and practice threat hunting with Kusto Query Language (KQL).

Objectives

- Deploy a Windows VM in Azure as a honeypot with RDP (port 3389) open to the internet.
- Ingest Windows Security Event logs into a Microsoft Sentinel workspace.
- Create an analytics rule to detect successful RDP logins (EventID 4624).
- Trigger and verify the alert using Microsoft Defender for Cloud.
- Harden the VM by blocking RDP access after observation.
- Hunt for real brute-force attack data using KQL (EventID 4625 — failed logins).

Tools & Services

Service	Role
Microsoft Azure	Cloud infrastructure — VM, NSG, networking
Microsoft Sentinel	SIEM — log ingestion, analytics rules, threat hunting
Microsoft Defender for Cloud	Alert management and security posture monitoring
KQL (Kusto Query Language)	Log analytics and threat hunting queries
Azure NSG (Network Security Group)	Firewall rules controlling RDP access

2. Environment Setup

2.1 Azure Resource Group

A dedicated resource group named **labtestvm_group** was created in the **South Africa North** region to contain all lab resources. A secondary NetworkWatcherRG was auto-provisioned by Azure for network diagnostics.

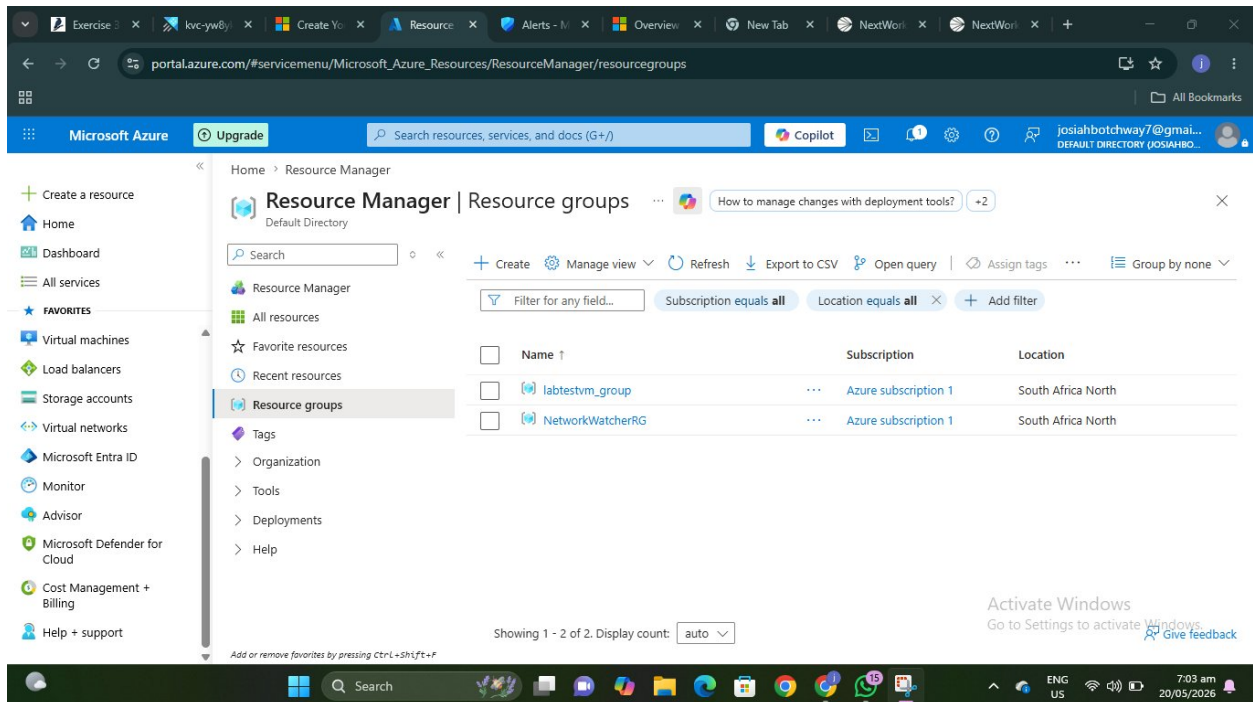


Figure 1 — Azure Resource Groups: labtestvm_group and NetworkWatcherRG

2.2 Virtual Machine Configuration

A Windows virtual machine named **labtestvm** was provisioned with the following specifications:

Property	Value
VM Name	labtestvm
Operating System	Windows Server
Size	Standard D2s v3 (2 vCPUs, 8 GiB RAM)
Region	South Africa North (Zone 1)
Public IP	4.222.185.235
Virtual Network	labtestvm-vnet / default subnet

Resource Group

LABTESTVM_GROUP

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labtestvm virtual machine agent status is not ready. Troubleshoot the issue →

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Essentials

Resource group (move) LABTESTVM_GROUP

Status Stopped

Location South Africa North (Zone 1)

Subscription (move) Azure subscription 1

Subscription ID 1ff46148-9101-43c3-8cc2-bd6d444c39b1

Availability zone 1

Operating system Windows

Size Standard D2s v3 (2 vcpus, 8 GiB memory)

Primary NIC public IP 4.222.185.235

Virtual network/subnet labtestvm-vnet/default

DNS name Not configured

Health state

JSON View

Activate Windows Go to Settings to activate Windows.

Add or remove favorites by pressing Ctrl+Shift+F

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Figure 2 — labtestvm Overview: Windows VM in Stopped state with public IP assigned

3. Exposing RDP to the Internet (Honeypot Configuration)

To simulate an insecure cloud workload and attract real-world attack traffic, an inbound NSG rule was created to **Allow** all traffic on port 3389 (RDP) from any source. This is the exact misconfiguration that exposes thousands of real-world Azure VMs to brute-force attacks daily.

The rule was set at priority 300, ensuring it takes precedence over the default DenyAllInBound rule (priority 65500).

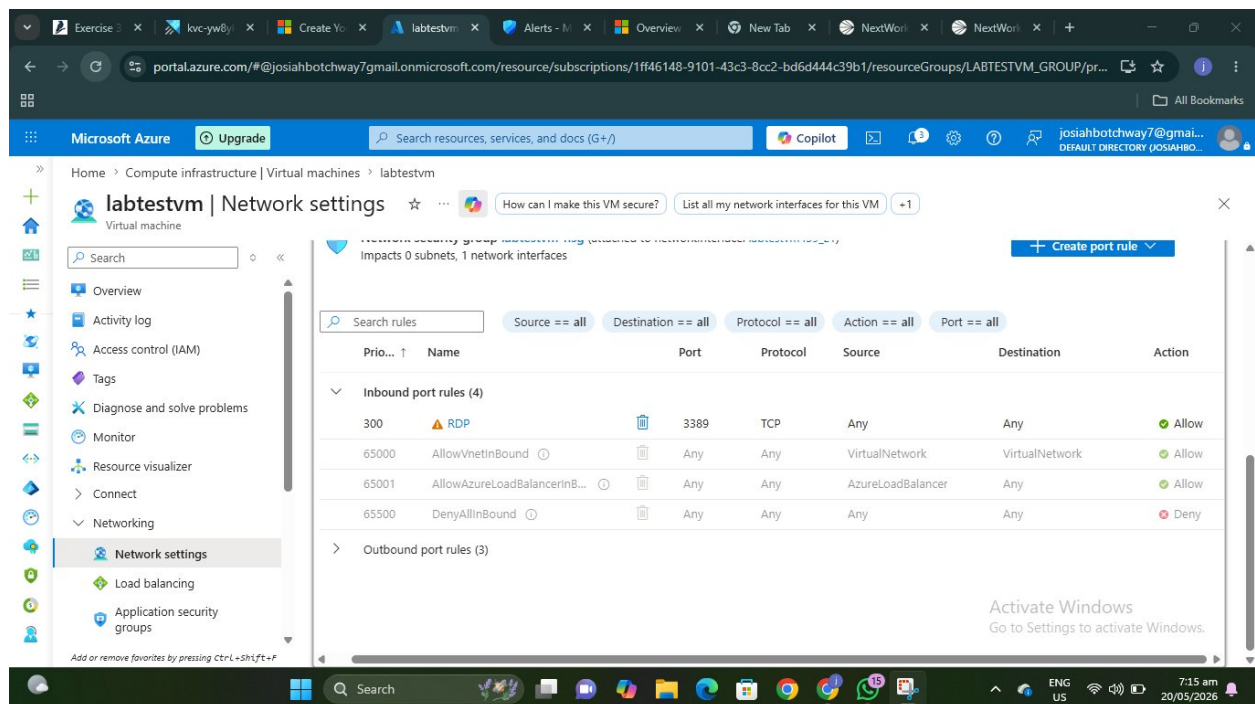


Figure 3 — NSG Rule: RDP port 3389 set to ALLOW from Any source (honeypot state)

■ **Security Note:** Exposing RDP port 3389 publicly is a critical misconfiguration in production environments. This was done intentionally in a lab setting to generate real attack telemetry for Sentinel analysis.

4. Microsoft Sentinel — Detection Rule & Alert

4.1 Sentinel Analytics Rule

A custom analytics rule was created in Microsoft Sentinel to detect successful RDP logins to the honeypot VM. The rule monitored for **Windows Security EventID 4624** (Successful Account Logon) filtered by logon type 10 (RemoteInteractive / RDP). When triggered, it generated a high-severity incident visible in Microsoft Defender.

4.2 Simulated RDP Login

A test login was performed using Windows Remote Desktop Connection (mstsc) to the VM's public IP **4.222.185.235**. This simulated the action of an attacker who successfully gained access after a brute-force attempt, allowing validation of the detection pipeline.

4.3 Alert in Microsoft Defender

Within minutes of the test login, Microsoft Defender for Cloud raised a **High-severity alert: "Successful local sign-ins"** with status *New*. This confirmed the end-to-end detection chain — from log ingestion through Sentinel analytics to Defender alert — was functioning correctly.

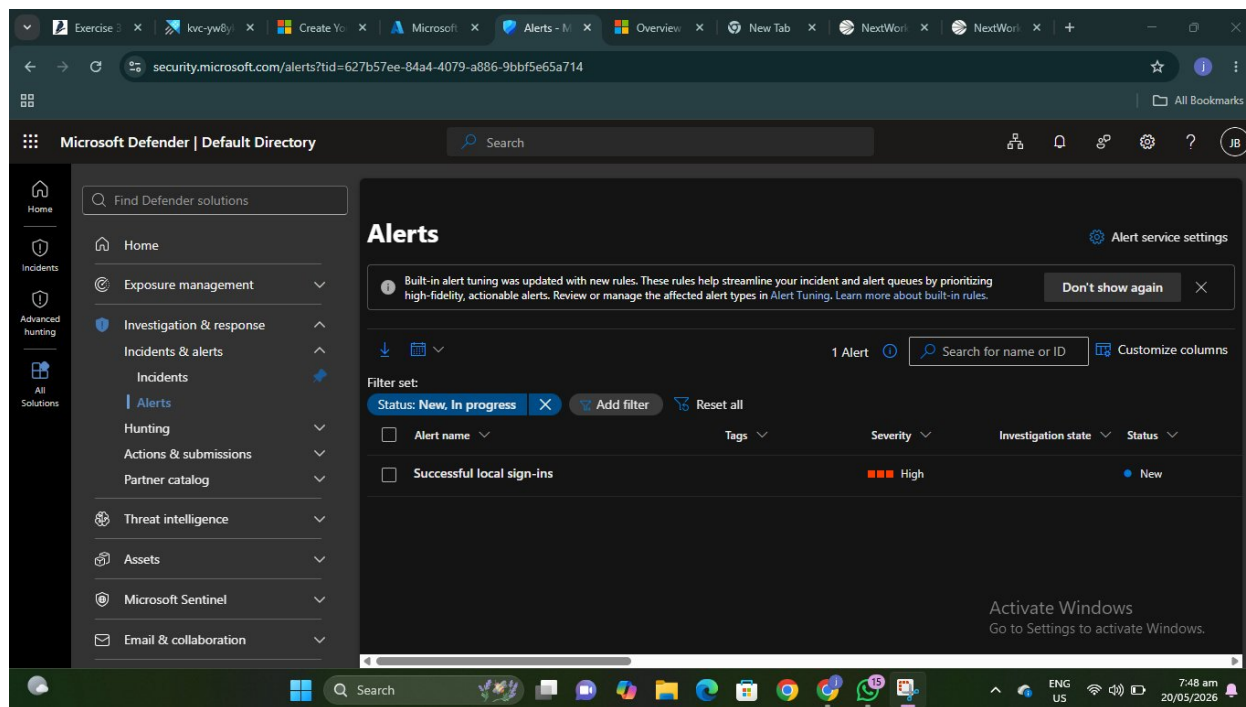


Figure 4 — Microsoft Defender Alert: 'Successful local sign-ins' (High severity, New status)

5. Hardening — Closing RDP Access

After confirming the detection rule was working, the NSG inbound rule for RDP was updated from **Allow** to **Deny**. This immediately cut off all public RDP access to the VM, simulating an incident response action — closing the attack surface after the threat has been identified.

The rule retained the same priority (300) and source/destination (Any), with only the Action changed to Deny, making the transition clean and auditable.

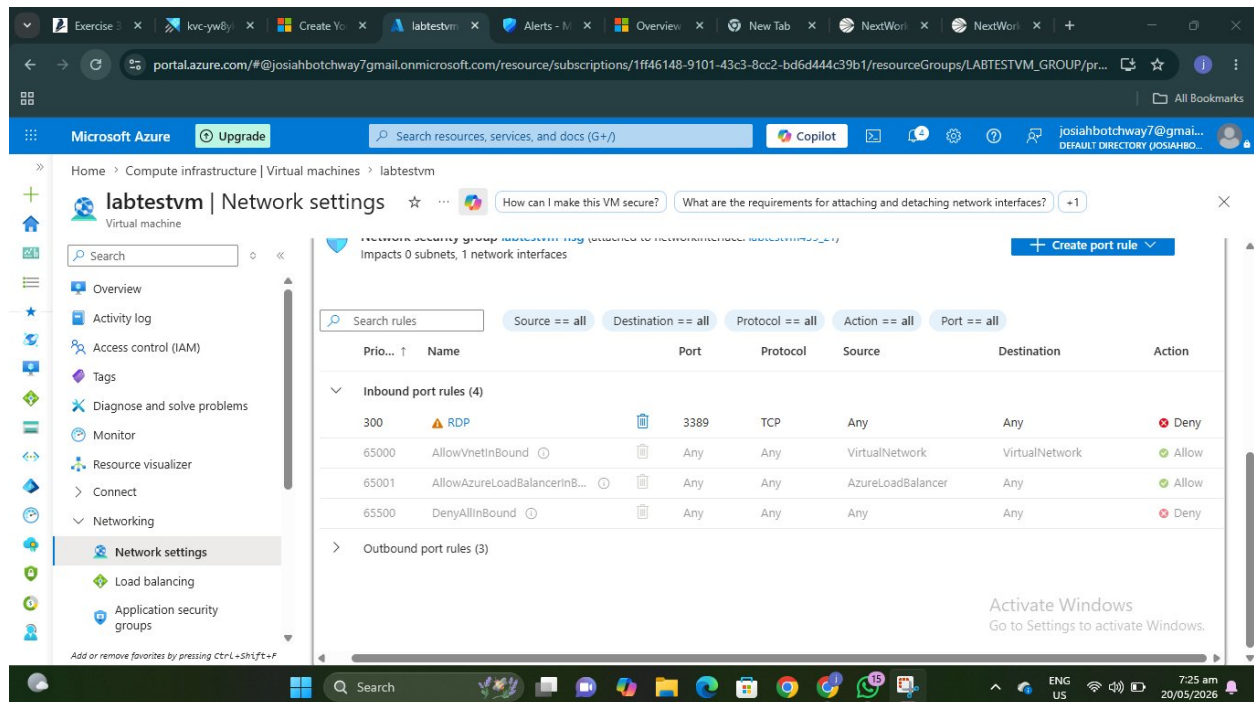


Figure 5 — NSG Rule Updated: RDP port 3389 changed to DENY (hardened state)

6. Threat Hunting — Brute-Force Attack Analysis

With the honeypot active, real-world attackers began targeting the exposed RDP port within hours. Microsoft Sentinel's Log Analytics workspace captured thousands of **Windows Security EventID 4625** (Failed Logon) events from multiple external IP addresses — evidence of active brute-force attacks.

6.1 KQL Threat Hunting Query

The following KQL query was used in Sentinel Logs to surface the top attacking IPs by failed login volume over the past 24 hours:

```
SecurityEvent | where TimeGenerated > ago(1d) | where EventID == 4625 | summarize  
Count=count() by IPAddress, Computer | where Count >= 15 | order by Count desc
```

6.2 Results — Top Attacking IPs

The query returned significant brute-force activity. The top offending IP addresses each generated over 9,900 failed login attempts against **labtestvm** within a single 24-hour window:

IP Address	Target VM	Failed Login Count
185.156.73.74	labtestvm	10,214
185.156.73.169	labtestvm	10,133
92.63.197.69	labtestvm	10,010
185.156.73.59	labtestvm	10,003
92.63.197.9	labtestvm	9,957
185.156.73.69	labtestvm	9,942

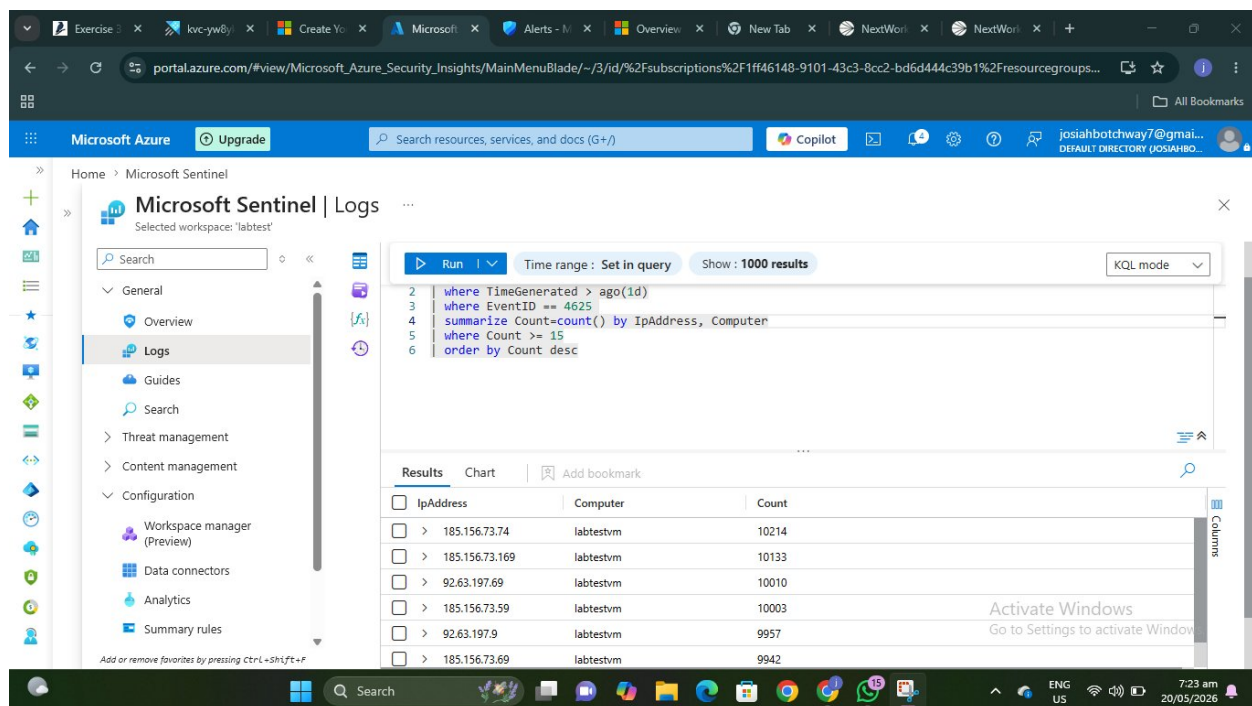


Figure 6 — Sentinel KQL Hunt: Top brute-force attacker IPs by failed login count (EventID 4625)

The 185.156.73.x subnet and 92.63.197.x range were consistently represented, suggesting coordinated or automated scanning activity from those network blocks. Over 10,000 attempts from a single IP in 24 hours is characteristic of automated credential-stuffing or dictionary-attack tooling.

7. Key Findings & Lessons Learned

Exposed RDP = Immediate Target: The VM attracted thousands of brute-force attempts within hours of port 3389 being opened — demonstrating how rapidly threat actors scan cloud IP ranges.

Sentinel Detection Works End-to-End: The custom analytics rule correctly detected the simulated successful RDP login and surfaced a High-severity alert in Microsoft Defender, validating the full pipeline.

KQL is Powerful for Threat Hunting: A simple 6-line KQL query was sufficient to identify the top attacking IPs, count failed logins, and rank threats by severity — a core SOC analyst skill.

NSG Rules as First Line of Defence: Changing the NSG rule from Allow to Deny immediately hardened the VM. In a real incident, this would be an immediate containment action.

Free-Tier Honeypots Attract Real Traffic: Even a low-powered VM with a public IP in an African Azure region received automated attack traffic from European IP ranges within the same day, confirming global internet scanning is continuous and indiscriminate.

7.1 Recommendations

- Never expose RDP (or SSH) directly to the public internet on production VMs — use Azure Bastion or a VPN gateway instead.
- Enable Just-in-Time (JIT) VM access in Microsoft Defender for Cloud to limit RDP exposure to approved time windows.
- Use strong, unique passwords and enforce MFA for all VM logins.
- Configure geo-blocking NSG rules to restrict RDP to known corporate IP ranges.
- Enable Microsoft Sentinel analytics rules for failed login thresholds to detect brute-force activity proactively.
- Regularly review Sentinel workbooks and hunting queries for signs of lateral movement post-compromise.

8. GitHub Repository Structure

This project is published as a portfolio piece on GitHub. The repository contains all documentation, KQL queries, and screenshots that make up this lab. Below is the recommended structure:

```
azure-soc-honeypot-lab/ ■■■■ README.md # Project overview & setup guide ■■■■ docs/ ■
■■■■ Azure_SOC_Honeypot_Project.pdf # This document ■■■■ screenshots/ ■ ■■■■
resource-group_snapshot.png ■ ■■■■ vm-overview_snapshot.png ■ ■■■■
rdp-open_snapshot.png ■ ■■■■ rdplogin-success_snapshot.png ■ ■■■■
rdp-closed_snapshot.png ■ ■■■■ brute-force-hunt_snapshot.png ■■■■ kql/ ■ ■■■■
brute_force_hunt.kql # EventID 4625 hunting query ■ ■■■■ rdp_success_detection.kql #
Sentinel analytics rule query ■■■■ LICENSE
```

8.1 Suggested README Sections

- Project Overview — what the lab does and why
- Architecture Diagram — VM, NSG, Sentinel workspace flow
- Prerequisites — Azure subscription, Sentinel workspace setup
- Step-by-Step Lab Guide — from VM provisioning to KQL hunting
- Key Findings — brute-force stats and detection results
- KQL Queries — copy-pasteable hunting and detection queries
- Screenshots — annotated evidence of each phase
- Lessons Learned & Recommendations

This document was produced as part of a hands-on SOC skills project. All attack data is real, captured from internet-facing honeypot infrastructure on Azure. No production systems were affected.